

TAEUK IM

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INTERESTS

Orbital Edge Computing, Satellite Constellation Design, Space Data Centers, CubeSat Mission Design, Satellite Networks

EDUCATION

M.S. in Aerospace Engineering

KAIST (Korea Advanced Institute of Science and Technology), Daejeon, South Korea

Sep 2025 - Aug 2027 (Expected)

GPA 3.66 / 4.3 (3.67 / 4.0)

- Advisor: Prof. Jihwan Choi | ACAI Laboratory (Aerospace Communications & Applied Intelligence Lab, <https://acai-kaist.github.io>)

B.S. in Astronomy

Yonsei University, Seoul, South Korea

Mar 2023 - Aug 2025

GPA 3.88 / 4.3 (3.72 / 4.0)

- HIGH HONORS, 1ST SEMESTER, 2024; HONORS, 2ND SEMESTER, 2023

Undergraduate studies in Electrical Engineering (prior to transfer)

Sejong University, Seoul, South Korea

Mar 2018 - Dec 2020

GPA 4.38 / 4.5 (3.95 / 4.0)

CONFERENCES/PRESENTATIONS

T. Im, S. Kim, I. Choi, and J. Choi, "Performance Evaluation of Space Data Centers: DDSSO-Assisted LEO Mega-Constellations," *The Korean Institute of Communications and Information Sciences (KICS) Summer Conference 2026*, Jun. 2026.

T. Im and J. Choi, "Multi-Shell Space Data Centers in the Sun-Synchronous Orbit-Assisted LEO Mega-Constellation," *The Korean Institute of Communications and Information Sciences (KICS) Winter Conference 2026*, Feb. 2026.

MANUSCRIPTS UNDER REVIEW

1. I. Choi, S. Kim, T. Im, and J. P. Choi, "Dynamic Virtual Network Embedding for Online Task Offloading in LEO Mega-Constellations," submitted to *IEEE Transactions on Mobile Computing*, 2026.

RESEARCH EXPERIENCE

Graduate Research Assistant

KAIST | Daejeon, Korea

Sep 2025 - Present

Investigating orbital edge computing architectures that combine conventional LEO mega-constellations with dawn–dusk sun-synchronous orbit (DDSSO) nodes and heterogeneous onboard GPUs to address power and computational bottlenecks from an astrodynamics and space-systems perspective.

- Developing a multidisciplinary design optimization framework that links spacecraft volume, solar-array size, orbit-dependent instantaneous and average solar power generation, and onboard GPU capability; evaluating whether higher-performance computing on DDSSO nodes can provide a better TSR–CAPEX tradeoff than expanding an LEO-only constellation. (Current)
- Formulated a system model and proposed an evaluation metric for the DDSSO-assisted LEO space data center; designed and propagated LEO and DDSSO orbits in MATLAB and quantified QoS gains over a conventional LEO-only constellation while evaluating the effects of DDSSO altitude and onboard computing capability; paper submitted to KICS.
- Proposed a DDSSO-assisted LEO architecture that extends satellites beyond conventional bent-pipes to orbital edge computing and space data center nodes; proposed the architecture leveraging the near-continuous solar illumination of DDSSO plane to support higher onboard computing capability and submitted a position paper to KICS.
- Developed a numerical two-body orbit propagator in MATLAB using ODE45. Verified the conservation of orbital quantities and cross-checked propagated states against an independent Keplerian propagator and GMAT's RK89 integrator across different orbit cases; extended the framework to include J2 perturbation and am currently developing a TudatPy-based orbit propagation and solar-power generation framework to evaluate orbit-dependent power availability.

Undergraduate Research Assistant (Advisor: Prof. Jae Kyu Suhr)

Sejong University | Seoul, Korea

Sep 2019 - Feb 2020

- Studied fundamentals of digital image processing, including edges, gradients, filters and transforms
- Learned core concepts of computer vision, including least squares, RANSAC and Viola-Jones object detection
- Studied basics of deep neural networks (DNNs), including CNN-based image classification and YOLO-based object detection

PROJECTS

Conceptual Design of a Reusable Unmanned Space Vehicle

Sep 2024 - Dec 2024

- Defined an Earth-observation mission concept for a reusable unmanned space vehicle and led the GNC subsystem, designing the mission sequence from direct LEO insertion and on-orbit imaging through fuel-efficient deorbit, atmospheric reentry, and runway recovery for reuse.
- Designed and simulated a circular sun-synchronous repeat-ground-track operational orbit in MATLAB and GMAT for repeated daytime Earth observation, incorporating J2-induced nodal regression; analyzed altitude and inclination trades and verified multi-day ground-track repeatability.
- Coordinated with subsystem leads to translate payload, power, communications, propulsion, and structural constraints into the operational orbit, orbit maintenance concept, navigation architecture, and deorbit plan.
- Developed an orbit-design framework to maximize the J2-induced RAAN precession rate for faster global coverage, combining MATLAB optimization with polynomial fitting to reduce repeated computation.

KOREA CANSAT COMPETITION

Jul 2024

- Led the design of a vision-based CanSat mission to estimate the Sun's position from camera-derived elevation and azimuth, transforming observer-centered coordinates into Earth-centered inertial coordinates; extended the concept to general object identification and coordinate estimation

- Designed the CanSat sensing and onboard-processing architecture, integrating dual Raspberry Pi cameras, GPS/IMU, magnetometer, and barometric sensing with passive camera stabilization and onboard solar detection; developed system and algorithm block diagrams for the complete data flow.
- Led the Electrical Power System (EPS) and Command & Data Handling (C&DH) subsystem design, defining the battery and power-distribution concept and onboard processing flow; coordinated subsystem interfaces and overall system integration as the team lead.

Literature Review on Computation Offloading in SAGIN: Architecture, Optimization, and DRL Approaches

Sep 2025 - Dec 2025

- Reviewed recent Space-Air-Ground Integrated Network (SAGIN) architectures for distributed computation, focusing on how tasks are placed and processed across user devices, aerial platforms, LEO satellites, and terrestrial cloud servers; compared binary and partial offloading, multi-tier execution paths, and heterogeneous computing roles across network layers.
- Analyzed optimization and deep reinforcement learning approaches for computation offloading and resource allocation in dynamic SAGIN environments, with emphasis on latency–energy tradeoffs, computing-resource constraints, and adaptive decision making across heterogeneous edge nodes.

Computational Astronomy

Mar 2024 - Jun 2024

- Implemented and evaluated core numerical methods in Python for scientific computing, including interpolation and Fourier-based signal analysis, ODE integration, and nonlinear equation solving and optimization; validated custom implementations through analytical solutions, convergence and error studies, and NumPy/SciPy comparisons.
- Applied numerical methods to reconstruct sparsely sampled supernova light curves, perform 2-D interpolation and Fourier-domain analysis of galaxy images, and fit nonlinear dark-matter density models to observational data; evaluated model accuracy and parameter uncertainty using chi-square and Hessian-based analyses.
- Evaluated numerical methods under varying sampling rates, step sizes, initial guesses, and boundary conditions; analyzed convergence, numerical error, and accuracy–computation tradeoffs to identify appropriate algorithm settings for different problems.

SKILLS

Technical: Python, MATLAB, GMAT, C++, Arduino, LaTeX, Machine learning, Linux

WORK EXPERIENCE

Government-Funded Teaching Assistant

Sep 2026 - Present

KAIST | Daejeon, Korea

- Supporting the Korea–Poland International Defense Cooperation Program.

Teaching Assistant

KAIST | Daejeon, Korea

- KAIST-Saudi Space Professional Development Program - KAIST Space Institute (Summer 2026; Developed MATLAB modules for Walker Delta constellation modeling; KAIST ground station visibility and feasible satellite analysis; Satellite link budget and data rate evaluation)
- Advanced English Listening and Advanced English Reading (Summer 2026; Prof. Josalyn Knapic)
- Reading Great Books on Human Intelligence and Civilization: Universe (Spring 2026; Prof. Myung-Hyun Rhee)

Teaching Assistant

Korea University (KU) camp

- Taught students in drone assembly and flight practice, Arduino practice, and coding exercises (Summer 2025)

Teaching Assistant

Korea Astronomy Olympiad (KAO)

- Instructed students in Astronomy Olympiad problem-solving and astrodynamics, and supervised telescope observations (Summer 2023; Summer 2024; Winter 2024)

Social Service Agent (Military Duty)

Jan 2021 - Oct 2022

Jongno Police Station

TEST SCORES

TOEFL | 96/120 | Dec 2025

AWARDS & HONORS

Government-sponsored scholarship

Sep 2025

KAIST | Daejeon, Korea

- Government-sponsored scholarship recipient during the Master's degree program at KAIST

Full Scholarship

Yonsei University | Seoul, Korea

- Granted full scholarship for the semester in recognition of outstanding academic performance (Fall 2024)

Commendation Award (Certificate of Appreciation)

Oct 2022

Jongno Police Station | Seoul, Korea

Full Scholarship

Sejong University | Seoul, Korea

- Full scholarship awarded for first-place academic performance (Spring 2019; Spring 2020)